5.1 Basic concepts of Tissue Culture

Basic concepts of plant tissue culture are totipotency, differentiation, dedifferentiation and redifferentiation.

Totipotency

The property of live plant cells that they have the genetic potential when cultured in nutrient medium to give rise to a complete individual plant.

Differentiation

The process of biochemical and structural changes by which cells become specialized in form and function.

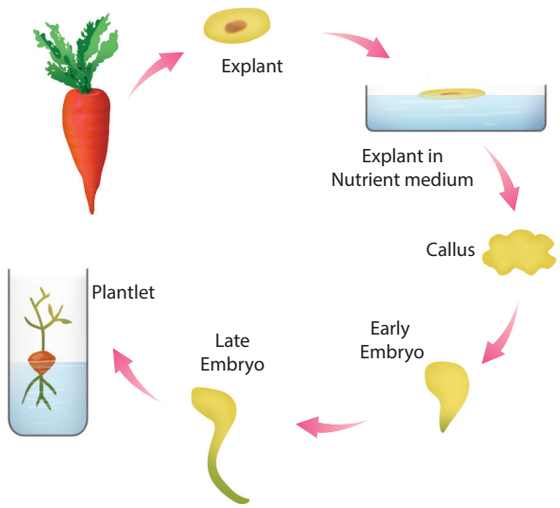


Figure 5.1: Totipotency

Redifferentiation

The further differentiation of already differentiated cell into another type of cell. For example, when the component cells of callus have the ability to form a whole plant in a nutrient medium, the phenomenon is called redifferentiation.

Dedifferentiation

The phenomenon of the reversion of mature cells to the meristematic state leading to the formation of callus is called dedifferentiation.

These two phenomena of redifferentiation and dedifferentiation are the inherent capacities of living plant cells or tissue. This is described as totipotency.

5.2 Plant Tissue Culture (PTC)

Plant tissue culture is used to describe the *in vitro* and aseptic growth of any plant part on a tissue culture medium. This technology is based on three fundamental principles:

- The plant part or explant must be selected and isolated from the rest of plant body.
- The explant must be maintained in controlled physically (environmental) and chemically defined (nutrient medium) conditions.

Explant: The tissue taken from a selected plant transferred to a culture medium often to establish a new plant.

5.2.1 Laboratory Facilities for PTC

For PTC, the laboratory must have the following facilities:



Figure 5.2: Tissue culture lab

- Washing facility for glassware and ovens for drying glassware.
- Medium preparation room with autoclave, electronic balance and pH meter.
- Transfer area sterile room with laminar air-flow bench and a positive pressure ventilation unit called High Efficiency Particulate Air (HEPA) filter to maintain aseptic condition.
- Culture facility: Growing the explant inoculated into culture tubes at 22-28° C with illumination of light 2400 lux, with a photoperiod of 8-16 hours and a relative humidity of about 60%.

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